

Yifei Chen

Senior software engineer

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PROFILE

Senior Software Engineer with 10+ years of experience in scalable web application development using React, TypeScript, React Native and Node.js. Delivered large-scale features for Meta AI, Twitter Spaces, and hybrid office platforms, impacting hundreds of millions of users. Known for building intuitive interfaces, robust services, and collaborating cross-functionally to deliver measurable product outcomes. Thrives in performance-critical, user-centered environments across AI, real-time systems, and productivity tools.

SKILLS

Frontend — React.js, Next.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Redux Toolkit, React Query, Headless UI, ARIA (Accessibility), SVG, Canvas, WebGL, WebSocket, WebRTC, Chart.js, D3.js

Backend — Node.js, Express.js, NestJS, TypeScript (Back-end), REST API Design, Webhooks, OAuth 2.0, Stripe API, Redis, PostgreSQL, MongoDB, JSON Schema, JWT Auth

AI & ML Integration — PyTorch, Transformers.js, TensorFlow.js, Llama 3.2 API integration, Vision Transformers, multimodal inference pipelines

Testing & QA — Jest, React Testing Library, Cypress, Playwright, Supertest, Postman, Pact (contract testing), axe-core (ally testing), BrowserStack

PROFESSIONAL EXPERIENCE

Meta, Senior Software Engineer

01/2024 – present

- Spearheaded the frontend development of the **Meta AI App** (Llama 3.2 Integration) using React and TypeScript, delivering real-time multimodal AI interactions to 400M+ users.
- Engineered a streaming chat interface with intelligent message rendering and photo/audio support, achieving a 98% Lighthouse performance score.
- Introduced image editing features using WebGL + Canvas APIs and natural language commands powered by Llama 3.2, increasing engagement on visual tasks by 34%.
- Connected Node.js backend services to Meta's model APIs with batch routing and retry strategies that reduced latency by 22%.
- Designed fallback workflows and optimistic UI for voice-to-text responses using browser audio APIs and async token streams.
- Devised test harnesses for visual response validation, maintaining 91% unit test coverage and catching 89% of regressions pre-production.
- Collaborated with AI infrastructure to deploy services via Kubernetes and maintain 99.98% uptime across all supported platforms.
- Redacted PII in photos using in-browser Vision Transformer filters; ensured GDPR/CCPA compliance.
- Offered product feedback and debugging support across 3 internal toolchains: Horizon SDK, PyTorch inference API, and Meta Design System.
- Presented new interaction flows in company-wide AI summit; proposals adopted in April 2025 app refresh.

Twitter (now X), Senior Software Engineer

02/2016 – 12/2023

- Optimized the **Twitter Spaces** web experience using React, TypeScript, Next.js, Redux Toolkit, enabling browser-based joining/speaking in live audio rooms.
- Integrated full-stack audio event streaming with WebRTC, Node.js (Socket.IO), Redis Pub/Sub, reducing join latency to 150ms across 1M+ active sessions.
- Orchestrated a feature-rich scheduling/reminder UI using React Calendar, Luxon, and integrated iCal export + tweet promotion for Space visibility.
- Deployed monetized Ticketed Spaces via Stripe Connect API, building secure payment/auth flows in Node.js/TypeScript backend; directly contributed to \$3M+ revenue.
- Executed lazy loading and SSR for SEO optimization using Next.js dynamic imports and Headless UI, reducing initial paint time by ~65%.
- Embedded live metrics with Datadog RUM, internal Kafka logging streams, and built event replay dashboards using React + Chart.js.
- Partnered with Android/iOS engineers for platform parity and unified state handling using protobuf schemas and cross-platform interface contracts.
- Delivered closed-caption support and ARIA-labeled components with React ARIA, ensuring WCAG 2.1 AA accessibility for Spaces on web.

Robin Powered, Software Engineer

11/2014 – 01/2016

- Developed the **Web-Based Desk & Room Booking Platform** using React, Redux, SVG.js, and real-time state management with Socket.IO, enabling users to visually book desks in 3 seconds.
- Built room availability sync with Google Calendar API, Microsoft Graph API, and normalized recurring events in Node.js + PostgreSQL, improving booking accuracy by 98%.
- Developed interactive analytics dashboard using D3.js, Chart.js, and React Hooks, giving workplace managers actionable office usage insights.
- Authored Cypress E2E tests for all booking flows, set up browser matrix testing on BrowserStack, and reached 85% coverage.
- Prevented double-bookings with atomic update locks and optimistic concurrency control using Node.js and PostgreSQL advisory locks.
- Led usability sessions with product and UX, iterating UI via React Hot Loader and conducting A/B tests with LaunchDarkly.

Farseer, Inc, Full Stack Developer

01/2014 – 10/2014

- Assembled the **Web & Mobile Food Discovery Platform** using Node.js (Express), MongoDB, Bootstrap, and jQuery, launching on both desktop and mobile browsers.
- Enabled image upload with cropping and filter options using FilePond, Cloudinary API, and added location tagging via HTML5 Geolocation API.
- Established REST APIs secured with JWT-based OAuth2, role-based access control middleware, and password recovery flows via SendGrid.
- Devised moderation tools with AngularJS + REST hooks, supporting content reports and usage dashboards.
- Deployed app to AWS EC2, configured Nginx as reverse proxy, and added daily S3 backups + SSL using Let's Encrypt.

EDUCATION

Brandeis University, Master's degree in Computer Science

2012 – 2014

Tongji University, Bachelor's degree in Computer Science

2008 – 2012